

Altman Z-Score and Piotroski F-Score

Financial-strength factors and the distress anomaly

Group 25 | Mirko Balli and Lorenzo Attanasio
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HOW TO USE THIS GUIDE

This guide turns the slides into words you can say out loud.

Read the four short parts first - the story, the data, the method, and the three sentences at the end. They answer most of the questions. Then use the slide-by-slide part as your script. The **blue boxes** marked SAY THIS are written to be spoken almost word for word. Under each slide, IF THEY ASK lists the likely questions with a short, ready answer. **Green boxes** marked KEY MOVE are your strongest lines - the ones that win marks.

PART 1

The story in 60 seconds

This is the spine of the whole talk. Open with it.

SAY THIS (read this out loud)

We study two quality factors from the Global Factor Data of Jensen, Kelly and Pedersen. The first is the Altman Z-Score, from 1968. It measures the **level** of financial health: how far a firm is from bankruptcy. The second is the Piotroski F-Score, from 2000. It measures the **direction** of the fundamentals: is the firm getting better?

The common instinct is contrarian: buy firms in trouble and wait for the bounce. But the distress anomaly, from Campbell, Hilscher and Szilagyi, and from Dichev, says the opposite: distressed firms tend to underperform. So our question is simple - does paying for financial strength actually pay off?

Our results, one line each:

1. The Z-Score has **no premium** ($t = 0.21$). Safe firms and distressed firms returned about the same. This is the distress anomaly.
2. The F-Score **has a premium** ($t = 3.28$). But it rewards "improving fundamentals", not "cheap distress".
3. The two factors are **negatively correlated** (-0.36): level versus direction.
4. The F-Score premium **is profitability**. It disappears under the q5 model, where the ROE factor absorbs it.
5. **Only q5** prices both factors together (the GRS test). The CAPM, FF3, FF5 and FF6 do not.

6. Combining the two adds **diversification, not return.**

PUNCHLINE TO SAY

We do not just measure alpha - we explain where it comes from. The value of the F-Score is profitability, the ROE factor of q5. The Z-Score on its own is an anti-value bet with no premium.

The data, in plain words

Everything is monthly and for the United States, merged by year-and-month into one table. There are five sources.

1. Factors and portfolios - jkpfactors.com (Global Factor Data)

They arrive as zipped CSV files. We read the long-short returns of the Z-Score, the F-Score and the Quality cluster, plus two market returns, value-weighted and equal-weighted. These returns are already in decimals and already **excess**: each factor is long the high-score firms and short the low-score firms, so the position pays for itself. We also read the tercile portfolios - the top third and the bottom third of each factor - which we use to split a factor into its two legs (slide 5).

2. q5 factors - global-q.org (Hou, Mo, Xue and Zhang, 2021)

These are the market, size, investment, profitability (ROE) and expected growth, plus the risk-free rate. The file is in percent, so we divide by one hundred.

3. Fama-French factors - Ken French Data Library (downloaded in the script)

FF3 is market, size (SMB) and value (HML). FF5 adds profitability (RMW) and investment (CMA). We also take Momentum, and we call FF5 plus Momentum the FF6 model. The file is in percent, so we divide by one hundred.

KEY MOVE

We download these in the script so the work is reproducible: anyone who runs the code gets the data, with no extra files - only an internet connection on the first run.

4. Benchmarks - a Bloomberg terminal

The S&P 500 total return, MSCI USA Quality, MSCI USA Value, the US Aggregate Bond index, gold, and the US high-yield credit spread (the OAS). These are index levels, so we turn them into simple monthly returns. The one exception is the high-yield spread, which is already a level that measures stress in the credit market, so we keep it as a level. To compare these long-only benchmarks with our zero-cost factors, we put them in **excess returns** by subtracting the risk-free rate. The Bloomberg data is licensed, so it never goes into a public repository.

5. Macro variables - FRED (St. Louis Fed)

Financial conditions, the term spread, the NBER recession indicator, and the VIX. We use them only for the regime chart (slide 6).

SAMPLE SIZES (good to know)

Our two factors do not start at the same time. The Z-Score data start in **May 1962** (764 months), and the F-Score data start in **November 1952** (878 months). The Quality cluster goes back to 1927, which matters for one outlier (slide 4).

Q. Why only the United States?

A. The Global Factor Data covers many regions, but we chose the US for the long history and the data quality, and because the models we compare against - Fama-French and q5 - are built on the US market. We wanted the factor and the model to match.

The method, in plain words

If they ask about a method, here is a short answer you can say.

Zero-cost and excess return

A long-short factor buys one basket and sells another, so it funds itself. Its return is already an excess return - we do not subtract the risk-free rate. The benchmarks are long-only, so for a fair comparison we do subtract it.

Annualizing

Monthly mean times twelve gives the annual return. Monthly standard deviation times the square root of twelve gives the annual volatility. The Sharpe ratio is the monthly mean divided by the monthly standard deviation, times the square root of twelve.

Newey-West (HAC) standard errors

Monthly returns can be autocorrelated and can have changing variance. Plain OLS standard errors would be too small, so we use Newey-West standard errors, which are robust to both problems. We use them for every t-stat - both for the average returns and for the regression coefficients.

KEY MOVE

Say it out loud: all our t-stats are Newey-West, so they are conservative.

The t above 3 rule (Harvey, Liu and Zhu)

Hundreds of factors have been tested in the literature. With so many tests, the old t above 2 bar is too easy - this is the multiple-testing problem. Harvey, Liu and Zhu suggest about t above 3 for a believable new factor. The F-Score clears it at 3.28; the Z-Score does not, at 0.21.

Rolling windows (60 months)

We also compute the Sharpe ratio, the beta, the alpha and the correlation on rolling 60-month windows - five years. This shows that the premiums change over time; they are not constant.

Spanning regression

We regress a factor's return on a set of explaining factors (a model). If the intercept, the alpha, is clearly different from zero, the model does not explain the factor - there is return left over. If the alpha is about zero, the model explains or absorbs the factor. We read the alpha with a Newey-West t-stat.

GRS test (Gibbons, Ross and Shanken)

The GRS test is a **joint** test. The null says the alphas of all the test assets are zero at the same time. We test both factors, Z and F, together, for each model. If GRS rejects (a low p-value), the model leaves a joint alpha.

KEY MOVE

Only q5 is not rejected ($p = 0.32$), so only q5 prices both factors.

Internal spanning

We also regress F on Z, and Z on F, to see if one contains the other. F on Z leaves a significant alpha ($t = 3.44$), so F has return beyond Z. Z on F does not ($t = 1.29$). So **F subsumes Z**: F contains Z, not

the other way around.

Slide by slide - your speaking script

Slide 1 - Title

On the slide: title, authors, course, WU branding.

SAY THIS (read this out loud)

Good morning. We are Group 25 - Mirko Balli and Lorenzo Attanasio. Our topic is the Altman Z-Score and the Piotroski F-Score, two measures of financial strength, and the distress anomaly.

[Then go straight into the 60-second story from Part 1.]

Slide 2 - Factor Selection and Motivation (Part a)

On the slide: why these two factors. Z = level (five balance-sheet ratios), F = direction (nine accounting signals, 0 to 9). Both in the JKP Quality theme.

SAY THIS (read this out loud)

We picked two fundamentals-based quality factors. The Z-Score, from Altman in 1968, is a score built from five balance-sheet ratios. It tells you the **level** of health - how safe the firm is today. The F-Score, from Piotroski in 2000, is nine simple yes-or-no accounting signals, added up to a score from zero to nine. It tells you the **direction** - is the firm improving?

Both sit in the Quality theme of the dataset. They are the two pillars of Quality-minus-Junk, from Asness, Frazzini and Pedersen: the Z-Score stands for safety, the F-Score for fundamental strength. The contrarian instinct says buy the distressed firms and profit from the recovery. We test whether the market actually rewards that bet.

Q. What exactly does the Z-Score measure?

A. Altman 1968. A score from five balance-sheet ratios: working capital to assets, retained earnings to assets, EBIT to assets, equity to debt, and sales to assets. A higher score means a safer firm, further from bankruptcy.

Q. And the F-Score?

A. Piotroski 2000. Nine yes-or-no signals across three areas: profitability (ROA, cash flow, the change in ROA, accruals), leverage and liquidity (change in leverage, change in the current ratio, issuing new shares), and efficiency (change in margin, change in asset turnover). You add them up, zero to nine. A higher score means the fundamentals are improving.

Q. Why are both in the Quality theme?

A. They are the pillars of Quality-minus-Junk. Quality means profitable, growing and safe. The Z-Score is the safety side; the F-Score is the fundamental-strength side.

PLAN B

If you blank on the lists: "They are five balance-sheet ratios and nine yes-or-no signals. The key point is that the Z-Score is a static level and the F-Score is a change - an improvement."

Slide 3 - Interpretation and Relationship (Part b)

On the slide: both long-short; contrarian theory versus the distress anomaly; the rolling-correlation chart; three key correlations.

SAY THIS (read this out loud)

Both factors are long-short: long the high-score firms, short the low-score firms. The contrarian theory says distressed firms are riskier, so they should earn a premium. The distress anomaly says the opposite: distressed stocks systematically underperform - that is the work of Campbell, Hilscher and Szilagyi, and of Dichev. So the real edge is not "cheap", it is "improving" - which is exactly what the F-Score sorts on.

On the right, the rolling correlation between the two factors is mostly negative over time. Three numbers matter: Z and F are correlated minus 0.36, Z and Quality plus 0.56, and F and the market minus 0.37. So the F-Score is defensive, and the Z-Score moves with the Quality theme.

Q. Why are Z and F negatively correlated? Aren't they both quality?

A. **Key move:** they measure different things. The Z-Score is a level - a firm can already be safe, with a high Z, but no longer improving. The F-Score is a direction - a firm can be improving, with a high F, while starting from a low level. The firms with the highest level are often not the firms improving the most, so the signs go opposite ways.

Q. What does Z and Quality plus 0.56 mean?

A. The Z-Score moves with the Quality cluster, which makes sense - it is a quality measure, the level of soundness.

Q. What does F and market minus 0.37 mean?

A. The F-Score is defensive. In a downturn, the firms getting worse (the short leg) fall the most, so the short leg gains and the factor goes up when the market goes down.

Q. Is the correlation stable?

A. No - that is the point of the chart. It moves a lot, and even changes sign: it falls to about minus 0.87 around 2001 to 2002 and rises to about plus 0.63. More evidence that they capture different things.

PLAN B

A simple example: a stable utility has a high Z but an average F - safe but not improving. A turnaround firm has a high F but a still-low Z - improving but not yet safe. Different firms.

Slide 4 - Performance and Risk (Part c)

On the slide: Table 1 (Z, F, Quality, Market VW, Market EW) and the growth-of-one-dollar chart.

SAY THIS (read this out loud)

This table shows the risk and return of our factors over the full sample - all numbers are annual. Look at the Z-Score first: it returns almost nothing, 0.32 percent a year, and its t-stat is only 0.21. So the Z-Score premium is basically zero - we cannot tell it apart from luck.

Now the F-Score: it returns 3.23 percent a year with a t-stat of 3.28. That t-stat is above three, the modern bar from Harvey, Liu and Zhu for a factor we can trust. So the F-Score is real, the Z-Score is not. The chart on the right says the same thing: one dollar in the F-Score grows steadily, while the Z-Score line goes nowhere.

Factor	Return	Volatility	Sharpe	t (NW)
Z-Score	0.32%	10.99%	0.03	0.21
F-Score	3.23%	7.52%	0.43	3.28
Quality Cluster	2.55%	12.28%	0.21	2.88
Market VW	8.22%	18.38%	0.45	4.32
Market EW	11.26%	25.12%	0.45	4.05

Annualized, full sample. t is Newey-West. Quality is skewed by a 1932 outlier; on the common sample from 1962 its Sharpe is 0.60.

Q. Why does the Quality cluster have a Sharpe of only 0.21 but the chart grows nicely?

A. **Key move:** the full-sample numbers start in 1927, and the early years have thin coverage. One extreme month - August 1932, about minus 59 percent - dominates the statistics. On the common sample from 1962, the same start as our factors and the same as the chart, the Quality cluster has a Sharpe of 0.60 and normal kurtosis. We flag this in a note on purpose, to show we know the data.

Q. Why does the Z-Score line end below one dollar if its mean is positive?

A. Volatility drag. With a small positive average, about 0.32 percent, but 11 percent volatility, the compounded (geometric) return is slightly negative. Over sixty years that pushes one dollar below one. It is the gap between the arithmetic mean and the geometric mean.

Q. Newey-West versus OLS - what changes?

A. It corrects for autocorrelation and changing variance, so the t-stats are more conservative and more reliable.

Q. Why is equal-weight market higher than value-weight (11.3 versus 8.2 percent)?

A. Equal-weight puts more weight on small caps, so it earns the size premium but takes more risk - its volatility is 25 percent versus 18. The Sharpe is almost the same, about 0.45.

PLAN B

If asked for the t-stat of post-1962 Quality (not in the table): "The table reports the full sample; we quote the post-1962 Sharpe of 0.60 to match the chart. We can recompute the t-stat, but the point is that the 1932 outlier explains the difference."

Slide 5 - Where the Premium Comes From (Part c)

On the slide: each factor split into the long leg (top third) and the short leg (bottom third). Left chart = Z, right chart = F.

SAY THIS (read this out loud)

Here we split each factor into its two legs, using the top third and the bottom third. For the Z-Score, on the left, the safe leg and the distressed leg grow almost the same. The spread is about zero, so there is no premium. This is the distress anomaly made visible: the distressed firms do not underperform enough to pay you.

For the F-Score, on the right, the strong leg clearly beats the weak leg. The gap is wide, so the premium is real.

Q. So distressed firms don't underperform? Doesn't that contradict the distress anomaly?

A. **Key move - the subtle point:** the distress anomaly is about *risk-adjusted* returns, holding other things equal. Here, on a raw basis, the distressed firms grow like the safe ones but with more risk - more volatility, deeper drawdowns. So once you adjust for risk, the bet on distress does not pay. The Z-Score long-short pulls out no premium.

Q. Why thirds and not tenths (deciles)?

A. Thirds give more diversified, more stable legs. Deciles would be more extreme but noisier. Thirds are the standard way to see where the return comes from.

PLAN B

Keep the message: Z = equal legs = no premium; F = a gap = a real premium. This sets up the regressions.

Slide 6 - Rolling and Regimes (Part c)

On the slide: rolling 60-month Sharpe for Z and F; and the same Sharpe with the high-yield spread and NBER recessions on top.

SAY THIS (read this out loud)

These charts show stability over time. On the left, the rolling five-year Sharpe of both factors. Both have long flat or negative stretches - they stop working for years at a time. On the right, we put the same Sharpe against the high-yield credit spread and the grey NBER recession bands.

The message is that factor premiums are time-varying, not constant. That is a key risk-management caveat for the strategy.

Q. What does the high-yield spread have to do with these factors?

A. The Z-Score is about distress; the high-yield spread is how the credit market prices distress. The overlay checks whether the factors react to credit-stress regimes.

Q. Why is the Sharpe sometimes negative?

A. In those five-year windows the factor returned less than zero - it stopped working. That is timing and regime risk.

Q. The spread is rescaled - isn't that misleading?

A. It is only for readability, to share one axis. What matters is the relative moves - the peaks are the 2008 and 2020 crises - not the absolute level.

PLAN B

The risk-management line: "the factor premium is time-varying, so it is not a free lunch - it has to be managed."

Slide 7 - Publication Decay (Part c)

On the slide: Table 3 - performance in three periods per factor: in-sample (the paper's own window), pre-publication (after the sample, before publication), and post-publication. Framework: McLean and Pontiff.

SAY THIS (read this out loud)

This tests publication decay. For each factor we look at three windows. The Z-Score paper is from 1968 with a sample up to 1965; the F-Score paper is from 2000 with a sample up to 1996.

The F-Score shows the classic pattern: in-sample the t-stat is 4.48, and after publication it falls to 1.18. That is the McLean and Pontiff decay - once an anomaly is published, investors trade on it and the premium shrinks, by about a third on average. The Z-Score is different: it was never significant out of sample. After publication the return is basically zero, with a t-stat of minus 0.09.

Factor	Period	Return	Sharpe	t (NW)
Z-Score	In-sample (to 1965)	4.05%	0.80	1.48
Z-Score	Pre-pub (1966-67)	7.07%	0.77	1.10
Z-Score	Post-pub (1968 on)	-0.15%	-0.01	-0.09
F-Score	In-sample (to 1996)	3.83%	0.74	4.48
F-Score	Pre-pub (1997-99)	0.91%	0.10	0.18
F-Score	Post-pub (2000 on)	2.48%	0.24	1.18

Annualized. Pre-pub = after the paper's sample, before publication. The Z window is short (the data start in 1962).

Q. What is publication decay?

A. McLean and Pontiff 2016. After an anomaly is published, investors exploit it and the premium falls - about a third on average. The F-Score shows it: the t-stat drops from 4.48 to 1.18.

Q. Why is the Z-Score pre-publication window so short and weak?

A. **Key move - honesty:** the JKP data start in 1962, but Altman's paper is from 1968 with a sample up to 1965, so our in-sample window is only about 44 months. The low t-stat is from too little data, not from real weakness. We say this in a note.

Q. Is the F-Score still tradable?

A. It is weaker after publication - a t-stat of 1.18, no longer above three - but the economic engine, profitability, is still there. In fact the q5 ROE factor captures it (next slide). So it is not dead, it is explained.

PLAN B

If they note the Z pre-pub looks high (7 percent): tiny sample, about 24 months, statistically meaningless (t of 1.10). Don't over-read it.

Slide 8 - Spanning Regressions (Part d) - the technical core

On the slide: Table 4 (annual alpha, t, R-squared for CAPM, FF3, q5 and FF6 on Z and F), the GRS box, and two beta insights. Slow down here.

SAY THIS (read this out loud)

This is the core of the analysis: do standard models price our factors? We regress each factor on four models and look at the alpha. Start with the Z-Score. Under the CAPM the alpha is basically zero. Under q5 it is also not significant. It only looks positive under FF3 and FF6, and I will explain why in a second.

Now the F-Score: under the CAPM and FF3 the alpha is large and very significant - above 4 percent, with t-stats near 5. But under q5 the alpha disappears: minus 0.8 percent, not significant. That is the key result. The reason is in the beta: the F-Score loads on the q5 profitability factor, with an ROE beta of 0.52 and a t-stat of 9.5. Profitability absorbs the premium. The Z-Score, instead, loads strongly negative on value - an HML beta of minus 0.78 - so it is really a growth-and-safety bet.

The joint GRS test agrees: the CAPM, FF3, FF5 and FF6 are all rejected, but q5 is not, with a p-value of 0.32. So only q5 prices both factors at the same time.

Factor	Model	Alpha (ann.)	t (NW)	R-sq.
Z-Score	CAPM	-0.9%	-0.50	6%
Z-Score	FF3	+2.7%	2.84	64%
Z-Score	q5	+1.8%	0.95	41%
Z-Score	FF6	+3.1%	2.99	68%
F-Score	CAPM	+4.7%	4.87	14%
F-Score	FF3	+4.4%	4.87	29%
F-Score	q5	-0.8%	-1.04	61%
F-Score	FF6	+1.2%	1.68	57%

Alpha annualized, t is Newey-West. FF5 is omitted (FF6 dominates and tells the same story) but is included in the GRS test. q5 = Hou, Mo, Xue and Zhang 2021.

GRS JOINT TEST

Null: both factors' alphas are zero, run per model on the pair {Z, F}. The CAPM, FF3, FF5 and FF6 are rejected (p below 0.01). **q5 is not rejected (p = 0.32)**. Only q5 prices both factors together.

Key betas: F on the q5 ROE factor = 0.52 (t = 9.5); Z on FF3 HML = -0.78 (t = -9.7). And for the record, the CAPM market betas are small: about +0.17 for Z (slightly pro-cyclical) and about -0.19 for F (defensive).

Q. What is a spanning regression?

A. We regress the factor on a model. If the alpha is significant, the model does not explain it. If the alpha is about zero, the model absorbs it.

Q. Why does q5 explain F but the others don't?

A. q5 has a profitability factor, ROE, that directly captures the F-Score's profitability signals. The ROE beta is 0.52 with a t-stat of 9.5, and the alpha collapses to minus 0.8 percent. FF5 also has a

profitability factor, RMW, so it reduces the alpha to about 2.4 percent - but it does not kill it. q5 kills it completely.

Q. What is the GRS test and why does only q5 pass?

A. **Key move:** it is a joint test on both factors at once - the null is that both alphas are zero. We run it per model. Only q5 is not rejected, $p = 0.32$, so only q5 prices Z and F together. The others leave a significant joint alpha.

Q. Why does the Z-Score have a positive alpha under FF3 and FF6 if it has no premium?

A. **Key move - the subtle trap:** the Z-Score on its own earns nothing. But under FF3 it loads very negative on value, an HML beta of minus 0.78. Value has a positive premium over history. The model removes that negative value bet, and what is left over - the intercept - turns positive. It is not a real, tradable premium: it comes from the anti-value exposure. With the right model, q5 or even the CAPM, the alpha is gone, and the GRS test catches it.

Q. What does "Z is anti-value" mean?

A. High-Z safe firms tend to be expensive growth and quality names; low-Z distressed firms are often cheap value. Long high-Z and short low-Z means being short value, so the HML beta is negative.

Q. FF6 versus FF5?

A. FF6 is FF5 plus Momentum. We include it as a fuller alternative model. We drop FF5 from the table because FF6 dominates and tells the same story; FF5 is still in the GRS test.

Q. What is q5 exactly, and why not q4?

A. q5 is Hou, Mo, Xue and Zhang 2021: market, size, investment, ROE and expected growth - the fifth factor. The earlier q4, from Hou, Xue and Zhang 2015, has no expected-growth factor. We use the five-factor version, so we cite the 2021 paper.

PLAN B

If you get lost in the numbers, hold three sentences: (1) the Z-Score has no real premium - it is anti-value; (2) the F-Score has a premium, but it is profitability - the q5 ROE absorbs it; (3) the GRS test: only q5 prices both.

Slide 9 - Conclusions (benchmark and combined factor)

On the slide: a bar chart of Sharpe ratios (Z, F, Combined RP, S&P 500, MSCI Quality, Bonds, Gold), four conclusions, and a long-only caveat.

SAY THIS (read this out loud)

To close, we compare the Sharpe ratios - all in excess returns, from 1989 - and we give four conclusions. First, the F-Score is a genuine premium, t of 3.3; the Z-Score is not, t of 0.2 - the distress anomaly in action. Second, level versus direction: the two factors are negatively correlated, minus 0.36. Third, the F-Score's edge is profitability, and only the q5 model prices both factors, GRS p of 0.32. Fourth, combining them adds diversification, not return.

One important caveat on the chart: the benchmarks are long-only indices, while our factors are long-short and zero-cost, so the Sharpe ratios are not directly comparable.

Series	Sharpe (excess, from 1989)
Z-Score	0.03
F-Score	0.43
Combined (risk-parity)	0.40
S&P; 500 TR	0.59
MSCI Quality	0.73
Bonds Agg	0.43
Gold	0.32

Benchmarks are long-only (excess of the risk-free rate); Z, F and the combination are long-short and zero-cost, so Sharpe ratios are not directly comparable. Combined 50/50 Sharpe 0.34 (t 2.62); risk-parity 0.40 (t 2.93).

Q. MSCI Quality has a Sharpe of 0.73, above the S&P at 0.59 and above your factors. So are the factors useless?

A. **Key move - the caveat:** no, because they are not comparable on risk. The benchmarks are long-only and carry the equity market premium; our factors are long-short and zero-cost and carry no market exposure. The value of the factors is their low correlation - they are diversifiers - not their standalone Sharpe.

Q. Why combine Z and F if the combination (0.40) does not beat F alone (0.43)?

A. Combining with the Z-Score, which has no premium, dilutes the F-Score. The benefit of the combination is diversification - lower volatility, around 5 percent versus 7.5, and smaller drawdowns - not higher return. Risk-parity does better than 50/50 because it underweights the high-volatility Z-Score.

Q. What is risk-parity, and did you avoid look-ahead?

A. **Key move:** the weights are inversely proportional to each factor's rolling 36-month volatility, so each contributes equal risk. We lag the weights by one month - we weight month t using only information up to month t minus one - so there is no look-ahead. This fix lowered the risk-parity Sharpe slightly, from 0.41 to 0.40. We report the honest number.

Q. Correlation with the benchmarks?

A. Close to zero with bonds, gold and equities, so they are real diversifiers.

PLAN B

The closing line: "the factors do not replace equities - they complete them, because they are uncorrelated."

Slide 10 - References

On the slide: 13 papers plus the data sources.

SAY THIS (read this out loud)

These are our references - thirteen papers and the data sources.

Q. Why cite Quality-minus-Junk (Asness et al. 2019)?

A. It is the anchor paper for quality investing. It defines quality as profitability, growth and safety. The Z-Score is the safety proxy and the F-Score the fundamental-strength proxy, which justifies our framing on slide 2.

Q. Why is Newey-West in the bibliography?

A. It is the method behind our standard errors (HAC). Citing it is the correct thing to do.

Q. And Gibbons, Ross and Shanken 1989?

A. It is the joint test on the alphas that we use on slide 8.

The hardest questions (cross-cutting)

Q. Do your results suffer from data mining or multiple testing?

A. That is why we use the Harvey-Liu-Zhu bar of t above 3. The F-Score clears it at 3.28, the Z-Score does not, at 0.21. And we did not search for a winner - we chose two factors in advance for an economic reason, level versus direction, and we report the null result for Z as well.

Q. Survivorship bias or look-ahead in the data?

A. The JKP factors are built point-in-time and include delisted firms - that is a standard feature of the dataset. On our side, the only possible look-ahead - the risk-parity weights - is lagged by one month.

Q. Transaction costs?

A. We do not model them - the factor returns are gross. We state this as a limitation. The F-Score has moderate turnover, since accounting signals update slowly. In practice costs would lower the premium, but they would not change its sign or its explanation, which is profitability.

Q. Why monthly?

A. It is the standard frequency in asset pricing, and it matches Fama-French and q5, which are also monthly. And the accounting signals update slowly anyway.

Q. Why capped value-weighted?

A. It is the standard JKP construction. It weights by market value but caps the mega-caps, so two or three names do not dominate the factor. It is more representative and more robust than pure equal-weight.

Q. Your sample includes the Great Depression and COVID - does that distort things?

A. For our two factors the sample starts in 1952 and 1962, so it does not include 1929 to 1932. The Quality cluster does go back to 1927, and that is exactly why we show the 1932 outlier and the post-1962 numbers.

Q. What is the practical take-away for a manager?

A. Three things. One: do not bet on cheap distress (the Z-Score) - it does not pay. Two: the F-Score pays, but it is essentially a profitability exposure (the q5 ROE), so you can get the same premium from known quality and profitability factors. Three: as low-correlation diversifiers they add value in a portfolio, but they are not a substitute for equities.

Q. If F is explained by q5, is it useless?

A. Not useless, but not incremental once a model already includes profitability. It is a clean asset-pricing result: the anomaly exists, but it is subsumed by a known risk factor, ROE. **Key move: this is a strength of our thesis, not a weakness.**

Q. Why do Z and F give opposite conclusions if both are quality?

A. Because quality has two dimensions: the level (Z) and the direction, or change (F). The market prices the second - improving fundamentals is informative and risky - but not the first - a static level of safety is already in prices. That is the heart of our contribution.

Advanced technical Q&A (only a deep examiner)

These come up only from an examiner who knows the data and methods well. The code is already correct and standard - do not change it. Knowing these answers is a top-mark move.

Q. The rolling alpha is annualized by times twelve - why not compounded?

A. Simple annualizing (monthly alpha times twelve) is standard in asset pricing. For alphas near zero, like ours, the difference from compounding is invisible. And it is only a chart, not a submitted number. No change needed.

Q. Which Newey-West lag or bandwidth did you use?

A. The automatic bandwidth from the Newey-West 1994 rule - note, not Andrews 1991, which is the one in kernHAC. On monthly data with about 760 observations it picks roughly four to six lags, and the results are robust to fixed lags of four to six. You can check the exact value with `bwNeweyWest`. No change needed.

Q. In the GRS test, why divide the residual covariance by T and not by T minus K minus 1?

A. Because the GRS test uses the maximum-likelihood estimate of the residual covariance, which divides by T, as in the original 1989 paper. The finite-sample correction is in the scale factor - T over N, times (T minus N minus K) over (T minus K minus 1) - not in the covariance denominator. Dividing by T minus K minus 1 would be wrong for the GRS formula. Do not change it.

Q. Fama-French uses its own market, q5 uses its own market - are the alphas comparable?

A. We use each model with its own market factor, as published by its authors. That is the correct practice in spanning studies - mixing them would be the mistake. The two markets correlate about 0.99, so they do not drive the conclusions: what kills the F-Score alpha is the q5 ROE factor, not the market. Do not change it.

Q. The rolling windows overlap - doesn't that create autocorrelation?

A. Yes, consecutive rolling estimates share data and are correlated. But we use them only to show the variation over time (a chart), not to do inference on the rolling values. The significance tests are on the full series, with Newey-West standard errors, not on the windows. So the overlap is not a problem here.

THE GOLDEN RULE

Changing correct code to "look more rigorous" lowers the mark. Rigor is shown by knowing why a choice is right, not by adding complications. An expert examiner expects exactly these choices.

Presentation tips

Timing (8 to 10 minutes, about one slide per minute)

- Slides 1 to 3: about 3 minutes - story, motivation, relationship. This is where you convince. Open with the contrarian-versus-distress tension.
- Slides 4 to 7: about 4 minutes - the evidence: performance, decomposition, regimes, decay.
- Slide 8: about 1.5 minutes - the technical core. Slow down here.
- Slides 9 to 10: about 1.5 minutes - conclusions and close.

Who says what (a suggestion)

One person opens (slides 1 to 5 - the story and the evidence), the other closes (slides 6 to 10 - regimes, regressions, conclusions). Whoever presents slide 8 should be the strongest on the regressions.

Three sentences to know by heart

MEMORIZE THESE

1. The Z-Score measures the level of health, the F-Score the direction - that is why they are negatively correlated.
2. The Z-Score has no premium (the distress anomaly); the F-Score does, but it is profitability - it disappears under q5.
3. Only the q5 model prices both factors together (GRS $p = 0.32$).

The mistake to avoid

Do not say "we found a profitable anomaly". Say: "we tested two quality factors, found one significant (F) and explained its source - q5 profitability - and confirmed the distress anomaly for the other (Z)". Show rigor, not naive excitement.

Honesty earns marks

Mention the limits yourself: transaction costs not modeled, post-publication decay, long-only versus long-short, the 1932 Quality outlier. Examiners reward people who know the limits of their own work.

SUBMISSION REMINDER

One zip with three files: **25_code.R**, **25_data.RData**, and **25_slides.pdf**. Your files and slides already use Group 25, so just bundle them. If the number ever changes, rename the three files, then rebuild the .RData (it is git-ignored, so add it to the zip by hand). To rebuild it: delete the old .RData, make sure the Bloomberg file is named `bloomberg_benchmarks.xlsx`, and run the script once. Keep the Bloomberg file private.

Good luck. You know this material - the goal tomorrow is to sound calm and clear, not to recite numbers. Lead with the story, let the tables back you up.